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FISPACT-II
Worked example

Self shielding example

Self shielding is a next order correction for an inventory calculation.

Worked example: Irradiation of Tungsten for 10 years

1. Configure FISPACT-II to irradiate 1kg of Tungsten for 10 year
2. Use Tendl17, GEFY6.1 and the fluxes from the FNS getting started example

Irradiation of Tungsten

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- Irradiate Tungsten for 10 days

```
CLOBBER
MONITOR 1
NOFISS
<< -----collapse cross section data- >>
GETXS 1 709
<< -----condense decay data----- >>
GETDECAY 1
FISPACT
* transmutation run
 MASS 1.0  1
 W 100.0
<< -----irradiation phase----- >>
FLUX 3.0E+14
TIME 10 YEARS
ATOMS
END
* END OF RUN
```

```
# files file

#index of nuclides
ind_nuc /Users/fosterd/nuclear_data/TENDL2017data/tendl17_decay12_index


#Lib of cross sections
xs_endf /Users/fosterd/nuclear_data/TENDL2017data/tal2017-n/gxs-709


#fluxes
fluxes fluxes


# lib decay data
dk_endf /Users/fosterd/nuclear_data/decay/decay_2012

# collapsed cros sections data
collapxi COLLAPX
collapxo COLLAPX

# gamma attenuation data
absorp /Users/fosterd/nuclear_data/decay/abs_2012

# condenced decay and fiss data (in and out)
arrayx ARRAYX
```

```
1                                                  Summary Output

0            Time       Cumulative    Activity            Dose rate          Heat output        Ingestion dose       Inhalation dose       Tritium activity
            (step)       (Years)        (Bq)               (Sv/h)              (kW)                (Sv)                 (Sv)                  (Bq)
 -----Irradiation Phase-----
 Irradn     10.000 y      1.00E+01    1.30E+15            1.41E+04           3.93E-02            0.00E+00             0.00E+00              1.21E+10
```

Self shielding

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Within FISPACT-II self shielding is accounted for by altering the reaction cross-sections.

SSFCHOOSE *n nprint sym*

- *n* number of element or nuclides
- *nprint* is by default 0, in which case it prints the list of probability table data files and the nuclide mixture. If it is set to 1, then it additionally prints total and partial cross-sections and dilutions versus energy bin for all the nuclides to which self-shielding is being applied.
- *sym* may either be element names (e.g., Ti) or nuclide names (e.g., W182)

Self shielding

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PROBTABLE *m n*

- Causes the probability tables to be read for the nuclides specified by the elements listed by the SSFCHOOSE
- $m = 1$ apply self-shielding to the existing cross-section value as a multiplicative factor, $m = 0$ the cross-section values in the energy groups, for which there are probability table data, are replaced by values which are derived directly from the probability tables

Self shielding

SSFMASS *m indx*

- *m* total mass (kg)
- *indx*2 number of elements in the material to be used in the self-shielding calculation

prob_tab

- Need to point the files file to the probability tables

Irradiation of Tungsten with self shielding

```
<< -----condense decay data----- >>
GETDECAY 1
<< ---self-shielding correction-- >>
PROBTABLE 0 1
SSFCHOOSE 1 0
W
SSFMASS 1.0 1
W 100.0
FISPACT
* transmutation run
 MASS 1.0   1
 W 100.0
<< -----irradiation phase----- >>
FLUX 3.0E+14
TIME 10 YEARS
```

```
#index of nuclides
ind_nuc /Users/fosterd/nuclear_data/TENDL2017data/tendl17_decay12_index

#Lib of cross sections
xs_endf /Users/fosterd/nuclear_data/TENDL2017data/tal2017-n/gxs-709

#fluxes
fluxes fluxes

# lib decay data
dk_endf /Users/fosterd/nuclear_data/decay/decay_2012

# collapsed cros sections data
collapxi COLLAPX
collapxo COLLAPX

# gamma attenuation data
absorp /Users/fosterd/nuclear_data/decay/abs_2012

# condenced decay and fiss data (in and out)
arrayx ARRAYX

prob_tab /Users/fosterd/nuclear_data/TENDL2017data/tal2017-n/tp-709-294
```

```
1                                      Summary Output

0          Time      Cumulative   Activity        Dose rate     Heat output    Ingestion dose   Inhalation dose   Tritium activity
          (step)      (Years)       (Bq)            (Sv/h)          (kW)            (Sv)             (Sv)              (Bq)
 -----Irradiation Phase-----
 Irradn   10.000 y    1.00E+01    1.29E+15        1.37E+04       3.86E-02        0.00E+00         0.00E+00          1.21E+10
```

Comparison

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Normal

DOMINANT NUCLIDES

	NUCLIDE	ACTIVITY (Bq)	PERCENT ACTIVITY	NUCLIDE	HEAT (kW)	PERCENT HEAT	NUCLIDE	DOSE RATE (Sv/hr)	PERCENT DOSE RATE
	Total	1.2985E+15		Total	3.9330E-02		Total	1.4080E+04	
1	W 181	4.5485E+14	35.03E+00	Re184	9.4347E-03	23.99E+00	Re184	1.0625E+04	75.46E+00
2	W 185	3.7591E+14	28.95E+00	W 183m	7.8244E-03	19.89E+00	Ta182	1.3531E+03	96.10E-01
3	W 183m	1.5786E+14	12.16E+00	W 185	7.6401E-03	19.43E+00	W 187	9.0861E+02	64.53E-01
4	W 185m	1.5065E+14	11.60E+00	W 185m	4.7725E-03	12.13E+00	Re184m	5.6446E+02	40.09E-01

0	Time (step)	Cumulative (Years)	Activity (Bq)	Dose rate (Sv/h)	Heat output (kW)	Ingestion dose (Sv)	Inhalation dose (Sv)	Tritium activity (Bq)
-----Irradiation Phase-----								
Irradn	10.000 y	1.00E+01	1.30E+15	1.41E+04	3.93E-02	0.00E+00	0.00E+00	1.21E+10

Self shielding

DOMINANT NUCLIDES

	NUCLIDE	ACTIVITY (Bq)	PERCENT ACTIVITY	NUCLIDE	HEAT (kW)	PERCENT HEAT	NUCLIDE	DOSE RATE (Sv/hr)	PERCENT DOSE RATE
	Total	1.2935E+15		Total	3.8618E-02		Total	1.3697E+04	
1	W 181	4.5629E+14	35.28E+00	Re184	9.4210E-03	24.40E+00	Re184	1.0610E+04	77.46E+00
2	W 185	3.7582E+14	29.05E+00	W 183m	7.8135E-03	20.23E+00	Ta182	1.3558E+03	98.98E-01
3	W 183m	1.5764E+14	12.19E+00	W 185	7.6381E-03	19.78E+00	Re184m	5.6363E+02	41.15E-01
4	W 185m	1.5098E+14	11.67E+00	W 185m	4.7829E-03	12.39E+00	W 187	5.4827E+02	40.03E-01

0	Time (step)	Cumulative (Years)	Activity (Bq)	Dose rate (Sv/h)	Heat output (kW)	Ingestion dose (Sv)	Inhalation dose (Sv)	Tritium activity (Bq)
-----Irradiation Phase-----								
Irradn	10.000 y	1.00E+01	1.29E+15	1.37E+04	3.86E-02	0.00E+00	0.00E+00	1.21E+10

Self shielding conclusion

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- PROBTABLE
  - Causes the reading of the probability table
- SSFMASS
  - Defines the material to be used in the self shielding
- SSFCHOOSE
  - Defines the material composition
- prob_tab
  - location of the probability table

At present, the probability table data are available only for the 616 and 709 energy group structures for neutrons. Attempts to use this keyword with other projectiles or cross-section datasets with other than the 586, 616, 709 or 1102 energy group structure will cause Fispact-II to terminate with a fatal error.